

UNIVERSITY

**ADOPTION OF INFORMATION TECHNOLOGY IN GHANAIAN
HOSPITALS: A CASE OF THE 37 MILITARY HOSPITAL**

students

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Use section numbers for the entire work, such as

1.1 Background to the Study

2.1 Introduction

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Today, doctors, nurses and patients can benefit from the highest technology offered by Information and Communication Technology (ICT), which serves the overall operation of the hospital. The use of ICT in the services and operations of the hospital system is not quite common in Ghana (Boateng, 2018). The advance in ICT has made computer networks an important part of the economic infrastructure. This has led to hospitals that have embraced these changes to achieve a competitive edge. The actual challenge is how to make use of the Internet and other ICT applications to ensure that their potentials are used to create enhanced economic opportunities and to reduce deficiency. There is no doubt that when used in a creative sphere, ICT is most likely to advance the living standard of people by affecting different business and service sectors (Christensen, 2017).

One very common form of Health Information Technology is patient's self-care and education, including for example interactive websites and medical devices for self-monitoring (Christensen, 2017). Along with adapting ICT in health, the patients are being involved in more and more of their own healthcare processes (Rindfleisch, 2017). This also seems to be a welcome trend from the patient side, as more and more people are interested in their own health proactively looking for health information from the internet. Clearly, with

modern Information Technology also the communication between health professionals and patients and other actors can now be done faster than ever, even in real-time. Besides, one of the most commonly referred use of health IT is the electronic data storage and data sharing across providers. This also seems to be a welcomed trend from the patient side, as more and more people are interested in their own health and proactively looking for health information on the internet. Clearly, with modern Information Technology also the communication between health professionals and patients as well as other actors can now be done faster than ever, even in real-time. Besides, one of the most commonly referred use of health IT is the electronic data storage and data sharing across providers.

Several researches have heralded the potential of Information and Communication Technologies in health care in improving the productivity and quality of the services (Christensen & Remler, 2016). This is not surprising, since ICT has indeed helped to improve productivity in many fields of the economy. For instance, the banking sector has benefitted from the automation that was made possible by the new technology. Moving towards internet banking has reduced the need for labour that was tied up in the routines, and thus enhanced labour productivity. However, ICT has been estimated to contribute to the overall productivity growth by one percentage unit, implying that half of the whole labour growth comes from ICT (Pohjola, 2016). The health care industry is an information intense industry and could be expected to benefit from advanced Information and Communication Technologies. In addition to this, the issues of privacy have to be taken into account. All in all,

in health industry it is important to consider thoroughly before making any decisions, as mistakes and medical errors can have serious consequences that are not well tolerated by the society (For example, premature death) (Christensen, 2017).

1.2 Statement of the Problem

The use of health information and communications technology in improving productivity and overcoming the awaiting challenges has been widely acknowledged (Christensen & Remler, 2016). Therefore, to keep health services available to all, health care processes has to be improved. On the other hand, in the health care industry, adoption has been relatively slow. Health care is clearly an information intense sector which could benefit from information and communication technology, and the potential has also been noted.

The researches decided to pose some questions that need answers. Why is it taking so long to move towards the adoption of a more comprehensive health ICT? Furthermore, as an industry, ICT markets have some shortcomings that can act as barriers of adoption. Low product differentiation, high switching costs and the lack of technical compatibility are typical problems for ICT products and systems and can complicate health ICT adoption (Hillestad, 2015). Health clients are very heterogeneous in their needs and backgrounds, and the low product differentiation can become a problem. The switching costs can also be substantial, especially in big health organizations where huge amounts of information have to be transmitted into new formats etc. however, many problems impede the internal operations of IT departments concerning the accommodation of the patients and staff

transactions with respect to weakness in the management of data storage, network (wireless), communication, privacy and security, database, archiving, disaster recovery, multi systems interconnection, internet and intranet access to web and emails, etc (Haux, 2015). These problems form an impediment that decreases the performance and the quality of service in large hospitals. Moreover, users (doctors, nurses and technicians) in a fully digital hospital are facing many challenges such as how to deal with the system, how to enter and use data and how to find a comfortable interface between people and machines (Haux, 2015).

Quite a lot of studies have been undertaken on health information systems from the literature reviewed (e.g., Christensen, 2017; Manuelli, 2017). However, there is insufficient information in a development country context of Ghana. Therefore, it is necessary to understand the phenomenon from a developing context of Ghana.

Purpose of the Study

The main purpose of the study is to examine the adoption of ICT in Ghanaian hospitals using 37 Military Hospital as a case study.

Research Questions

The research questions are specifically to address the following issues:

- (i) What is the impact of Information Communication Technology on corporate revenue, corporate performance, customer service, and communication, quality and type of products?
- (ii) What factors affect ICT adoption at the 37 Military Hospital?

- (iii) What challenges hinder the adoption of ICT innovations at the 37 Military Hospital?

Objective of the Study

The study field is on Information Technology and the research area is on the adoption of Information Technology in Ghanaian Hospitals using the 37 Military Hospital as a case study.

General Objectives

The general objective of the proposed research is to add to the general body of information and research area of the adoption of Information Technology in Ghanaian hospitals using 37 Military as a case study. The research sought to bring to the consideration of hospital administrators, health care assistants, policy formulators and government of the essence of the adoption of Information Technology as the surest way of moving hospitals in Ghana to the digital health care services. The research focuses on how Information Technology provides a tool for better health care service delivery of Ghanaian hospitals.

Specific Objectives

To achieve the general objectives, the research was meant at addressing the following specific three objectives:

First of all, to examine the Information and Communication Technology on corporate revenue, customer service and communications, quality and type of products.

Also, to examine the factors affecting ICT adoption at the 37 Military Hospital and

Lastly, to identify the challenges that hinders the adoption of ICT innovations at the 37 Military hospital.

Significance of the Study

The findings of this study should inform management of the 37 Military hospital and other hospitals in Ghana to put into perspective a better understanding of the possible impact of ICT on the performance of hospitals in Ghana. The study will be significant because it will provide answers to the implementation of Information Technology at the 37 Military hospital. The information gathered will serve as a secondary data source for scholars that may wish to undertake further research.

Delimitation of the Study

The research study was carried out within the Accra Metropolis specifically the 37 Military hospital, located around the Jubilee House of Ghana. The project covered the usage of Information Technology in Ghanaian hospitals, case study the 37 Military hospital, Accra.

Limitations of the Study

Hardly will a research be concluded without any constraints. In a country such as ours where there is bureaucracy in the flow of information and where there is inaccurate data and statistics, there really has to be a problem when conducting a research works. Specific instances of constraints related to the data system were the cost of stationaries, printing

and transportation, inadequate data collection and handling procedure, lack of research skills and privacy of information considerations. Combining academic studies and the relatively short semester period was a major challenge. Some respondents were unwilling to give out details of information as well as procedures they employed in the hospitals with the suspicion of such information getting into the hands of the wrong people.

Organization of the Rest of the Study

The study is organized into five (5) chapters. The chapter one (1) which is the introductory chapter includes the background to the study, the research questions, statement of problem, significance and scope of the research work. Chapter two (2) focuses on different write ups of scholars which will be looked into, literature reviews, and various theories in the field of Internet Technology will be cited. Chapter three (3) focuses on methods used in carrying out the research. The fourth chapter deals with data analysis and findings of the study while the chapter five (5), which is the last provides a summary of the entire work, the conclusions and recommendations made for consideration.

CHAPTER TWO

REVIEW OF LITERATURE

2.1 Introduction

This chapter reviews the relevant literature with particular reference on previous works done by others in the area under study. It builds a theoretical foundation upon which the research is based. Furthermore, the review is meant to provide a foundation for the present work and serve as literature for further research.

2.2 Definition of Information and Communication Technology

Information and telecommunication technologies (ICT) can be defined as any technology used to facilitate processing, gathering, distribution and use of information (Beckinsale & Ram, 2016).

According to Nicole (2003), ICT is classified into information, telecommunication and networking technologies. ICT includes all forms of technologies such as internet, computers, websites, telephones, mobiles, wireless devices, networks (Manueli, Latu, & Koh, 2017) and hardware, software and telecommunication equipment (Kaiser, 2014).

Information and Communication Technology (ICT) in Health Care

The applications of information and communication technologies in medicine are commonly referred to as telemedicine and medical Informatics. Although these terms are often used together and confused with each other,

they are separate and have their own definitions. The institute of medicine (2016) defines telemedicine as the use of electronic information and communication technology to provide health care when distance separates the participants. It includes all forms of electronic communication between patients and providers and among providers, starting from telephone to interactive video and web-based communication. Medical Informatics is defined by The National Library of Medicine (2011) as the field of information science concerned with the analysis and dissemination of medical data through the application of computers to various aspects of health care and medicine. Medical Informatics can also be referred to as the intersection of information science, computer science and health care. For example, medical Informatics includes health care delivery processes that are supported by computers that help in analyzing electronic data (Christensen, 2017).

Christensen and Remler (2016) have roughly categorized the different possible application of ICT in chronic disease care in four groups: technologies that support a) patient self-care and education b) communication between patients and providers or between providers, c) electronic data storage and data sharing across providers, and d) the technologies that combine all these three applications. Although the authors have mainly discussed the case of chronic diseases, the results can in many respects be applied in other field of health care as well. Notably the application of the second and the third-group communication technologies and data storage- can be useful in other aspects of health care.

Self-Care and Education

Successful management of chronic disease care is facilitated considerably by active involvement of the patient in his / her own treatment procedure. There is also increasing willingness from the patient side to be integrated in their own health care process, and health care consumers are actively searching information independently (Detmer, 2013). The involvement is usually realized by patient education and information about his or her disease (Asmer, 2010) and information and communication technology can provide effective methods for patients' participation. This first category includes medical devices for self-monitoring as well as interactive website for education on the diseases. One example of self-monitoring equipment is software that accepts and analyzes data from blood glucose meters of people living with diabetes. As for interactive websites, especially in the United States there are many firms- mostly insurers- that maintain websites to help people with chronic diseases. Moving towards more self-care and patient and health consumer inclusion is largely associated with new ICT technologies and has been noted by other commentators as well as (Haux, 2016; Christensen, 2017).

Communication

All the communication technologies fall into the second group. It includes traditional telephone conversations, e-mails and videoconferencing. Modern communication technologies enable real-time communication and more frequent monitoring possible. Efficient communication plays a major role in adjusting care and preventing serious problems as interventions can be

done faster. Medical data can be transferred electronically in seconds and thus enables faster reacting to changes in patients' conditions (Christensen, 2017).

Electronic Data Storage and Sharing

The ICT applications in the third category- electronic data storage and data sharing across providers- have probably received the most attention. It has been stated that shifting from paper-based storing to Electronic Health Records is associated with remarkable cost-savings (Hillestad, 2015) and faster access to information, which results in improved efficiency. Also, unnecessary test can be avoided, when information can be easily found from the data base by different users. Electronic process also enables storing bigger quantities of medical data (Haux, 2015). This is essential as the amount and complexity of health- related information and knowledge constantly increases and has already made information processing a major component of any Health organization. Health ICT facilitates moving from decentralized and institution based towards more global data storing (Haux, 2015).

Having national health records can improve health care processes as different providers can access the same information fast and for example the duplication of test could be prevented. In the European Union the long-term goal is to have a system where all the clinicians in Europe can access health records from all countries (Andersen, 2016). This would improve conditions for treatment as the patient as well as the health care professional mobility is expected to increase (European commission, 2014). Without electronic records and communication technologies having wide databases would practically be impossible.

Combining Technologies

For instance, software that integrates and analyzes provider and self-monitored patient data combined with communication technology belongs to the fourth group. These technologies make it possible to do certain monitoring tests at home and send the data to healthcare professionals to be analyzed. When there is a need for intervention, it can be done inexpensively without delay. This kind of technical solutions have already been used in continuous remote clinical monitoring and have brought significant benefits to both patients and payers. For instance, in-care of hypertension patients, remote monitoring has helped to drop the blood pressure of the test groups (Artinian, 2011) and reduce the cost of the care (Christensen, 2017). There is also a steady increase of new technology such as Ubiquitous computing environment and sensor-based technology for health monitoring from distance (Haux, 2015). These technologies, coupled with modern communication technology to connect the patient and health professional belong to this group as well. These represent an area with a lot of potential for example in the home care of the elderly.

Slow Adoption

Although different form of information technology has been used in hospitals, for decades, the adoption of ICT has been relatively slow in the health care industry compared to many other industries in the banking industry for example, the ICT has made the automation of many routines possible and thus reducing the need for work force that was tied up in these routines. Consequently, the productivity of labour has increased permanently. However,

health care industry is lagging behind despite all the heralded benefits of ICT adoption (Shortliffe, 2015).

Lack of Demonstrated Cost-Effectiveness

One obvious barrier for new ICT adoption is the lack of demonstrated cost effectiveness of ICT investments in health care. As noted, before, there are many separate small-scale studies, but comprehensive studies have not been conducted yet. The decisions to adopt new technology are often irreversible, which means that it is extremely hard or impossible to go back to the old methods if the new technology proves to be poor. It is rational from the health care providers to be careful with their investment decisions as there is no considerable evidence of the benefits.

The Problems of Interacting ICT and Health Markets

Christensen and Remier (2016) have gone through some features of ICT industry and health industry that complicate the interaction between them. Health care industry differs in many aspects from other industries like banking, and these aspects can partly explain the slow adoption of ICT.

ICT Markets

ICT markets have some typical, significant shortcomings that can act as barriers of adoption in all markets. Christensen and Remler (2016) mention the low product differentiation, high switching costs and the lack of technical compatibility. When it comes to adopting new ICT, the lack of compatibility of all the different components of an ICT system can pose a major problem. New information and communications technology has to be interoperable with other electronic means of data storage and communications. One way of

overcoming this problem can be common technical standards. (Christensen, 2017) in the case of information and communications technology, it is also important to consider how well new technology interacts with the systems of other actors on the same field.

The so-called network *effect* means that the software or other forms of ICT must be interoperable with the systems of the other relevant providers or actors in the process. In our context, networks include both physical and virtual networks. Christensen and Remler (2016) give telephone and airline networks as examples of physical networks and networks of compatible fax machines or network of compatible software programmes for personal computers as examples of virtual networks, the value of a network depends on the number of existing users. For instance, as more and more people started using a fax machine, it became a more valuable way of communication. If there is only one person using the fax machine, it obviously has no value (katz, 2014).

It is also quite typical for the information and communications technology industry that product differentiation is low. The reason behind this, according to Christensen and Remler lies in the “positive feedback” mechanism related to the network effect. As the value of the network increases with the number of users it is likely that the bigger manufacturers with larger market share get even bigger and the smaller ones disappear altogether. As a result, it might happen that a single firm dominates the market when there is little diversity in the technologies available (Katz & Shapiro, 2014) have discussed the competition in the market with network effects, concentrating on the communication systems and hardware software markets. They have come

to the conclusion that positive feedback creates “tipping” which means that there is a tendency of one system to pull away from its rivals in its popularity as it has gained an initial edge. It follows that there is a tendency towards standardization as everyone starts using the same product.

There are fewer suppliers in the market, and thus less product differentiation (Christensen, 2017). Another typical feature for information and communications technology compared to other forms of technology is that the cost of adoption tends to be high. The reason for this is that in addition to the pure acquisition costs there will be plenty of other costs as well. First, the personnel must be trained frequently to use the new technology, which implies short-term costs as the time spent on learning takes time away from other tasks, which in turn decreases productivity. Moreover, switching from old paper records to new electronic records implies costs as all the existing information has to be transferred to the new system (Christensen, 2017).

Interaction between Health and ICT Markets

Christensen and Remler (2016) underscored that health care industry possesses some features that clearly intensify the general above-mentioned problems of ICT adoption. It is good to bear in mind that the paper by Christensen and Remler (2016) deals with the health care structure of the United States, which limits the possibilities of applying all that is said to other countries, including Ghana. First, there is a large number of actors affected by ICT decisions in health markets. The market structure is complicated with heterogenous patients, public, and out-of-pocket payers. Often as new communication technology is adopted, it will affect people in all stages of the service as each actor directly or indirectly depends on other actors replacing

one component of the communication or data sharing infrastructure causes switching costs to all interlinked agents, in the form on technological change or change in user requirements. To adopt new ICT, the network effect has to be so great, that it overcomes all the collective switching costs.

In the US this problem is intensified compared to other countries like Ghana since there are more agents involved in the process, particularly because of the bigger role of insurance markets. In some situations, also the patient is the user of the technology. Often, they tend to be older and not used to ICT, which increases the requirements for user-friendliness. One form of ICT will not fit all the patients' needs due to the heterogeneity. Platforms have to be adjusted for disease-specific cases, which imply more switching costs. (Christensen, 2017) there are in general three sources of switching costs. First, there are the costs of new hardware, such as mainframe computers and operating systems to store information. These costs also include the complementary products such as system management software to manage the information database.

New technology may also dictates the future hardware investments as the consumer must purchase products that are compatible with the old systems. The second source is the cost of information storage in databases and the cost of transferring information storage in databases and the cost of transferring information from the old databases to the new one. This can be very time consuming and costly, especially in big hospitals that have existed for a long period of time as there are huge amounts of information in the old formats that have to be moved to the new systems. There are also the old formats that have to be moved to the new systems. There is also the above-

mentioned risk of being locked-in to a specific system, if the information is in a specific non-transferrable format when even newer advanced information systems come up. Third there are the training costs that arise from teaching the personnel to operate with the new systems. These costs are the direct financial training costs as well as the costs that come from the lost time that could have been spent on other productive tasks (Christensen, 2017).

Switching to new ICT is costly and often risky. The risk of technical problem or break-down exists in all the industries and the financial damage can be substantial. However, in health industry also the societal costs of errors have to be taken into account. Medical errors can have serious and irreversible consequences at worst premature death. These kinds of outcome are not tolerated, and the decisions have to be done carefully (Manueli, 2017). As it has already come up, the consumers in health markets tend to be very heterogenous. They differ in their needs and services required. Compared to banking industry, the situation is very different. Many banking services can be automatic and similar to all the clients, such as paying the bills electronically. Health care is a very information intensive industry, and the services cannot be as standardized as in the banking example. When it comes to the actual care process, the provider needs to have all kinds of information of the patient that is not necessary for paying bills or other E-banking services. Even inside the same disease groups the needs differ based on the severity of the disease, age or socio-economic status. The characteristics of low product differentiation obviously limits the use of ICT (Christensen, 2017).

The issues of privacy, confidentiality and security also comes to mind in the context of ICT use in health care, especially in the case of databases.

Our medical records contain a lot of information about us such as height, weight, and blood pressure to mention some. Sometimes they contain very sensitive information about us that we do not want to be available to outsiders. Access to information about topics like fertility and abortions, sexual behaviours and psychiatric care must be controlled because disclosure can harm us. This kind of information can cause prejudice and embarrassment, complicate getting and holding a job affect our insurability. The issues of privacy, confidentiality and security have to be considered when adopting new ICT technology to store and exchange information (Rindfleisch, 2007).

Models for ICT Adoption

Sarosa and Zowghi (2013) believed that researches on ICT have so far only concentrated on the barriers and drivers of ICT adoption and most organizations are not on the adoption process due to the lack of adoption models, strategy and framework. Many researchers are of the idea that companies follow an ordered sequence to adopt ICT or new technologies. They start by basic application of the internet to fill integration of business systems and processes (Alonso & Fitzgerald, 2014; Spanos, 2012; Mehrtens, Cragg, & Mills, 2011; Rogers 2013). The process name changes in the firm's strategy, the firm's readiness, perceived benefits, external pressures on the firm, the firm's human capital and management and finally its openness to external sources of information, as convectional factors affecting ICT application by company's adoption. Grandon and Pearson (2014) found different ICT adoption factors in the literature, which, they summarized in to four main factors: external pressure on the organization, organizational readiness and its perceived ease of use and usefulness. On the other hand,

according to Davis (2006), external variables, usefulness, company attitude and ease of use are the main factors affecting the application of new technologies by organization.

Factors Affecting ICT Adoption by Organizations

According to Akkeren and Cavaye (2009) the characteristics of the firm and its owner/ manager and the cost/benefit analysis affect ICT adoption by organizations. Thatcher and Foster (2012) classify such factors affecting e-commerce and ICT adoption into two main categories: Inter Organizational (size, management and readiness) and Extra-Organizational (industrial, governmental and cultural). Price of technology, availability of networks, firm characteristics (size, age, ownership) and learning are quoted by Crespi (2014) as factors affecting ICT application. Competition levels and supplier-buyer relations are also among important factors (e-business watch report, 2018). In addition, Manuelli, 2017; Akkeren and Caveye (2009) have identified customer supplier/dependency, organizational readiness, structural sophistication of the firm, assertiveness, perceived benefits, perceived control, lack of time, computer literacy, subjective norm, mistrust of IT industry, external pressure to adopt, size, sector, status, the manager's age and experience and information intensity as factors affecting any ICT or e-commerce adoption decisions. The firm's whose managers and employees have ICT literacy, skill and expertise as well as access to internal and external support from ICT experts, will have more tendency to adopt ICT (Windrum & De Berranger, 2012). The age and experience of the manager/ owner and access to ICT experts within a business are also influential. Certain cultural traits, beliefs and values attached to resources, size of the firm and its social structures, nature of

social links and contracts may also influence ICT adoption in several different ways (Clayton, 2013).

According to some authors (Luque, 2010; Helpman, 2008; Breshnahan & Trajtenberg, 2015; & Dunne, 2014), ICT adoption, use and diffusion is directly affected by structural market characteristics (including the degree of competition and the nature of the firm's contractual relations with its suppliers and buyers), and individual firm characteristics (such as size and age). Small new plants are more likely to adopt advanced technologies than small old plants. The process of ICT adoption is affected by three (3) major factors technological (complexity, compatibility and relative advantage), organizational (management support, user resistance and level of expertise available), and environmental (supplier-customer relationship and competitive pressure) (Nakhai, 2018; Swatman 2008). Bandiera and Rasul (2012) maintain that firm's may be reluctant to adopt ICT due to their unfamiliarity with technology and lack of organizational readiness, including size, type, nature of business, perceived benefits. According to organization of economic operations development (2014) factors that possibly impact on ICT adoption by firm's are competition and rivalry, relationship with business partners, innovation, internal capacity, intra and inter-firm collaboration.

Benefits of Information and Communication Technologies

Poon and Swatman (2007) outline the perceived benefits of ICT/e-commerce into six (6) general classifications;

- a. Direct benefits: those which are quantifiable and can be measured by methods such as data analysis and by variables such as number of new customers.
- b. Indirect benefits: those which cannot be easily measured such as increase of customer loyalty and goodwill as a result of online services.
- c. Short term: the benefits that can be realized by companies within months.
- d. Long term: benefits such as additional customer enquiry and forming new networks. which may be achieved in long term and the exact time for their realization is not predictable.
- e. Tangible benefits: such as reduced production and administration costs, reduced lead time and increase sales.
- f. Intangible benefits: such as improved relations with business partners, improved internal; controls and quality of information.

Another classification was proposed by Boateng (2018), which the benefits are classified into 3 main groups; Strategic (including extending the firm's market reach, product differentiation, loyalty and retention of trading partners, improved revenue), Informational (including improved marketplace information, improved communication and relationships) and Operational (including reduced transactional costs and time efficiency). Yet another classification on a country level was provided by Boateng, (2018) in which the benefits of ICT are classified into three major groups; Economic (Increase in exports and trade, increase of Gross Domestic Product, employment opportunities, improve of local entrepreneurship), Socio-cultural (Education, Awareness creation, employee skills

development and improvement of gender development), Legal and others (Enactment of supporting policies, institutional development improving e-readiness).

Impact of ICT on Corporate Performance

There is no doubt that there are a lot of factors that influence the performance which include: the market structure in which a firm operates; the market share, size, human resources, managerial and organizational structure of the firm (Dunne, 2008). The impact of ICT adoption on corporate performance is subject to debate because the impact is not the same in different firms (Kohli & Devaraj, 2013; Chan, 2010). Some researchers believe that ICT has positive impact on business performance, (Stratopoulos, 2010; Barua, 2015), but others consider the impact insignificant (Yosri, 2012; Strassmann, 2010) or even negative (Setzekorn, 2008; Holland, 2017; Brynjolfsson, 2015). In anyway, if one us is to understand the impact of Information Communication Technologies on firm performance, he/she should consider information communication technology as an enabler of innovation (Koellinger, 2015).

Information Communication Technology may have positive impact on productivity on one firm and have less impact on profits in the same company (Brynjolfsson & Hitt, 2016), moreover, the benefits associated with ICT applications will be dependent on sector-specific effects and cannot be found equally in all sectors (Pilat & Wyckoff, 2013). Some information communication technologies may be influential on a firm performance than others for example computer networks may be

particularly important, because they enable a firm, to have closer relations with suppliers and buyers (Atrostic & Nguyen, 2012).

Impact of ICT on Corporative Turnover/Revenue

Positive impacts of ICT investments and usage on the firm's revenue have been proved in the health care sector (Devaraj & Kohli, 2013). While in the insurance industry, similar results were found. In this industry the insurers who had higher ICT expense ratio and lower non-ICT costs, experienced higher premium income growth (Harris & Katz, 1991). E-business Watch survey (2016) shows that 70% of firms who used ICT in the year 2016 in Europe, reported turnover growth. A research conducted on small and medium size enterprises SMEs in 14 African countries proved that ICT application had a significant positive effect on company's turnover (Esselaar, 2017; Khatibi, 2013; Windrtum & Berranger, 2012). According to UNCTAD report (2009), the firms that outpace their competitors in technological development are likely to capture market shares and profits from their competitors and even drive them out of business.

But profits from ICT enabled innovation are only sustainable until competitors are able to copy the innovation completely. Hence, clear relationship between ICT usage and profitability cannot be hypothesized because profitability is dependent on the competitive dynamics in a market which are difficult to observe in empirical studies. According to Vu (2014), in addition to the firm level revenue/ turnover, on a macro level, economic growth theories on country level predict that the economic growth is increased as a result of investing on ICTs. Various researchers

conducted on the impact of ICT on the economic growth indicate the contribution of ICT on economic growth indicate the contribution on economic growth has been very important for many countries. A research recently conducted by OECD proves that ICT is one of the key drivers of the rapid growth for economics, and the service industry is the main ICT user OECD (2011).

Impact of ICT on Corporate Market Share

ICT can help firms' cross boundaries of their markets, expand their market reach and as a result increase their share in the local and international markets (Pilat & Wyckoff, 2013; Windrum & Berranger, 2012; Kendall, 2011). In a study about the impact of ICT on Canadian firms, (Baldwin & Sabourin, 2012) found that a considerable amount of market share was transferred from firms who decline to use ICT to those firms which were increasing their ICT use. ICT will not only help firms avoid loss of market share but helps them increase such a share (Khongsin Tan, 2009; e-business watch report, 2008; OECD, 2014; Kendall, 2011). In addition, ICT and e-commerce can help firms reach narrow market segments which are scattered geographically (Turban & king, 2013; Turban, 2012).

Impact of ICT Innovation/Process Innovation

Many researchers show that there is a link between the use of ICT and the ability of a firm to innovate. This relationship was found by Bresnahan and Greenstein (2006). By testing the hypothesis that firms who introduce new products and processes or adjust their organizational structure can gain higher benefits from ICT investment than other firms,

Helpman (2008) prove that innovation and ICT are complementary. The literature service innovation is classified by Gallouj (2012) into three main categories: (i) Technological approach (taking into account the introduction and diffusion of new technologies into services that may improve productivity and performance;(ii)service oriented approach (believing that innovation in manufacturing and service industries are distinct); (iii) integrating approach (investigating the boundary between goods and services to develop a framework to bridge the dichotomous gap between them).

According to Miles, (2015), services are mostly interactive as they require close communication between service suppliers and their customers in most phases of service provision. As most service industries are information intensive, innovation focuses on this interaction as much as on conventional product and process features this capability enables ICT to change the costs, output and quality of most services. A study of OECD (2010) indicates that service sectors are major users of new technologies especially ICT nowadays ICT is considered as the most important technology for innovation in service sectors which helps them boost their economic performance (Licht, 2009). In addition to the impact on innovation itself, ICT can impact process innovation by facilitating inter-organizational links (Lee, 2010). In other words, ICT-enabled inter-organizational enhances innovation capabilities of firms by giving them the opportunity to share data, knowledge and resources and as a result, will optimize the value chain by eliminating the so-called bullwhip effect (European Commission, 2014).

Impact of ICT on Firm/Labour Productivity

There is a common agreement among most people that ICT has positive impacts on labor productivity and total factor productivity (Plait, 2015; Van Ark, 2012). It has been quoted by many researchers (E-business watch report 2008; UNCTAD, 2015; Walczuch, 2010) that ICT (especially computer) is not only an example of technological innovation, it also enhances firm's productivity. It is also estimated that network increases labour productivity by roughly 5 per cent (Pilat & Wycvkoff, 2013; Ark, 2012) but such an impact of ICT on firm and labour productivity vary significantly between sectors and among countries as full productivity growth is however only found when skilled labour is combined with organizational changes (European e-business watch report, 2008). A student of Canadian manufacturing companies showed that the plant with higher productivity growth were those who used greater number of advanced technologies (Baldwin, 2012). Based on OECD report (2014) between the years 1988 to 1997, the profitability and productivity of users of advanced technologies grew more than those of non-ICT users. But recent studies suggest that the positive impact of ICT (including productivity) may become stronger overtime (Eurostat, 2008).

Many researches indicate that ICT-related productivity growth is more observed in those sectors that use ICT more than other sectors. These sectors may include trade, financial services, business services, and the ICT manufacturing sectors themselves (Van Ark, 2012; Nordhaus, 2012).

Impact of ICT on Costs

Many researchers including (E-business watch, 2013; Nienhaus, 2013; Subramani, 2013; Cloete, 2012; Brown, 2011), have tried to show how ICT application can have positive impacts on the firm's cost efficiency. To them, cost minimization decisions are one of the main areas of a company's efficiency improvement via using modern technology. New ICTs have helped firms reduce the cost of gathering, storing, retrieving, processing, analyzing and transmitting information (Parham, 2014), transaction cost (Turban & King, 2013; Turban, 2012; Walczuch, 2010), and communication and information transmission cost (Ratnasingam & Pavlou, 2013; Bedi, 2009; Poon & Swatman, 2007). Quayle (2012) believes that in addition administration and production costs reduction, ICT can reduce lead time and stock and improve marketing and quality of information. Other costs reduce by ICT applications are: inventory cost (Turban & king, 2013), marketing costs (Poon & Swatman, 2007), operating and sale and purchase costs (Khong Sin Tan, 2009; Kendall, 2011).

Impact of ICT on Marketing Methods

Poon & Stawman (2007) believe that ICT can help firm introduce new method of direct or indirect marketing. In addition to introducing new methods of marketing, ICT can improve marketing and marketing techniques (Quayle, 2012).

Challenges of ICT Application

The barriers before ICT application have been classified in different general categories by many researchers. For example, Stockdale & Standing

(2014) group these barriers into five different categories including lack of resources, knowledge, skills, security concerns and firm's readiness. Education, management time, economic concerns and technical know-how has been raised (Cragg & king, 2008). Hadjimonolis (2009) classifies them into two general type: external (supply, demand and environmental) and internal (resources and system barriers).

Turban (2012) classify the ICT and e-commerce barriers and limitations in to technical (which can be solved by money including security, reliability and telecommunications, software, integration of e-commerce software with existing databases) and non-technical (which a more difficult to solve including people's attitude, lack or trust, resistance to change, faceless transactions). Based on the general categories proposed by different researchers for barriers for ICT adoption, for ease of reference, these barriers are classified into 4 main categories, namely Financial, Security and Technical, Managerial (Levy, 2015).

Implementation Costs

Cost is a double-edged sword, which can be a driver and an obstacle of new technologies/ecommerce adoption. ICT can lead to more cost-effective operations but the initial financial resources for adoption of ICT are a major concern (Hadjimonolis, 2009). In other words, initial investment for hardware, software, connection, maintenance, training and etc, is a major concern for firms to adopt e-commerce (Quaddus & Achjari, 2015). The implementation cost of ICT/e-commerce and lack of financial resources has a barrier have also been raised by other researchers including (Khong Sin Tan, 2009; UNCTAD, 2014; OECD, 2014; Stockdale & standing, 2014).

Limited Financial Resources

As mentioned, the adoption of new technologies requires initial heavy investments to purchase needs software's, hard wares, equipment, connections and to train staff etc. With regards to especially, small and medium sized enterprises, these companies seem to be limited in financial resources which Will impede them from obtaining the equipment to adopt new ICT, therefore, limited financial resources can be considered an important barrier of ICT adoption (Haqdjimanolis, 2009).

Concerns about Confidentiality

Security issues are still the main concerns of people that will impede them from transacting online. Most people have heard stories about wrong use of personal information therefore they doubt that sending all personal data through internet is secured (Miller & Weekert, 2010). In an online transaction, the type of the information required and the nature of the company which collects the information in one hand and the individual differences between customers on the other hand are the main sources of privacy concern for customers. Such privacy concerns can be grouped into five levels:

Collection (where the customers become sensitive about the huge amount of information required online), Unauthorized secondary use of Data (personal data collected may be used for another purpose without permission), Errors (accidental alteration of information which will corrupt the reliability of the data base), improper access (unauthorized access to personal data), invasion (unwanted contacts with consumers) Smith and Burke (1995). System errors invasion of customers privacy and unauthorized secondary data

use are the main concerns of customer about confidentiality which may prevent them from using any online application or new ICTs (Khatibi, 2013; Turban & King, 2013; Turban, 2012; Kendall, 2011).

Concerns about Fraud, Technical Faults

An online transaction may require an online payment by the customer, as a result of which the account and credit bank details of the buyer will be spread via internet. Even if the customer is convinced to provide his private and personal information doing online payment remains a major concern. An online business has always its own security risks which will concern customers. Some of these risks are of technical nature such as viruses which may invade the site during an online transaction (Tan, 2009) .in a survey conducted by United Nation Committee of trade and Development (UNCTAD) in the year 2014, 71 percent of all countries named security concerns (including viruses) as to be the main barriers impeding the use of ICT and ecommerce by customers.

Technical Barriers

To handle all processes of ICT application from designing web to online delivery of goods and services, firms may face with technical barriers which may impede them from ICT adoption. These technical barriers can be of two major types (Akkeren & Cavaye, 2010).

Complexity of New Technologies which Require New Skills

Every new technology seems to be very complex to use in the beginning firms may not have the necessary skill and expertise to apply such technologies and they have to either develop new skills inside the firm or

outsource part of the task. To acquire new skills in house or out sourcing it would be expensive for firms (Quayle, 2012) in addition to existing information systems of firms may be not proper for future development (Akkeren & Cavaye, 2010). Yet another problem exists if firms want to buy new information systems. Firms cannot trust the external IS suppliers (Hadjimonolis, 2009).

Limited in-House IT Skills

As mentioned, in order for firms to adopt ICT in their processes, they will not only need necessary technical facilities, but also, they would need technical staffs to work with new technologies. The lack of talented and skilled IT staff in firms to develop and maintain the e-commerce system and difficulties in finding and recruiting qualified employees level of IT knowledge of personnel, lack of technical resources and lack of computer and e-business knowledge among staff, have been all mentioned by different researchers as the main technical barriers of ICT \E-commerce adoption by firms (Khong Sin Tan, 2009). In addition, Turban and King (2013) and Turban (2012) mention other technical barriers for ICT and e-commerce adoption: some internet protocols not being standardized, certain processes are not reliable insufficient bandwidth, expensive software's, difficulty in integrating digital and no digital sales and production information, limitations of dial-up, cable, ISDN, wireless systems.

Managerial Barriers

Some of the barriers that halt the adoption of ICT in firms might be of managerial nature and relate to the management of a firm which can be classified into:

Insufficient Time Spent on Planning

In fact, the top management of a firm or its owner is the sole decision maker in a firm. With regards to ICT adoption, one of the barriers is that the management of the firm does not spend sufficient time on the matter and plan for that. The reason behind this might be that he is not aware of the importance of ICT adoption and the benefits that might be brought about for his firm (Turban & King, 2013).

Insufficient knowledge or Experience of Internet Services

It is obvious that in case the management of a company suffers from insufficient knowledge or experience of information systems or IT as well as managerial skill, he might resist against any adoption of new technologies or be reluctant to network with other enterprises (Jones, 2013).

CHAPTER THREE

METHODOLOGY

Introduction

This chapter explains how the research was undertaken. It deals with the techniques and procedures used in collecting data for the study. The structure of the chapter is presented as follows; research design, population, sampling and sampling procedure, instruments, data collection procedures, and data analysis.

Research Design

In this study, survey research design was adopted where data was collected from the employees of the 37 Military Hospital. The survey research strategy is the most popular and common strategy for social research including business disciplines (Saunders, 2009). This kind of research design was the most appropriate since it offered the researchers the chance to collect a large amount of data from a substantial population at a very low cost.

The data can be easily compared and analyzed using various statistical techniques. Survey design is usually the preferred research design for researchers who are interested in collecting original data to describe a population that is too large to observe directly. Careful probability sampling provides a group of respondents whose characteristics may be taken to reflect those of the larger population and carefully constructed standardized

questionnaires are not the only data collection technique in the survey research, but the questionnaire remains the most commonly used tool in survey.

This kind of research design was the most suitable since it will give the researchers the opportunity to collect primary data from the hospital. The choice of the survey design was motivated by the fact that it best describes the opinions, personal characteristics, perceptions, preferences and attitudes of respondents. The choice of a survey method enables a wider coverage area while giving respondents the opportunity to analyze their individual contributions and identify different barriers existing among the various population sub-groups.

Population

The population for the study was employees of the 37 Military Hospital in Accra. The population of the study was staff of the 37 Military Hospital including the HR manager and staff who carry out other administrative duties at the hospital. An assessment of the hospital shows that, the total number of staffs is about seven hundred and fifty (750) including casual workers.

Sample and Sampling Procedure

The sample size of the study was a total of one hundred and twenty (120) respondents. This number was used to ensure that there was a general presentation of the population. This is because the research is supposed to examine the adoption of information Technology in Ghanaian hospitals, case the 37 Military Hospital. This sample size included staff, nurses, doctors and

casual workers. The basis for the selection of these respondents was to get wide ranging thoughts and suggestions from respondents.

To help collect the relevant information as regards the topic under investigation, the sampling technique that was employed in selecting the sampling was convenience sampling technique. A convenience sampling technique was used for the selected fifty (120) respondents. Convenience sampling method is used in surveys where the researcher is interested in getting an inexpensive estimate of the fact. Convenience sampling was used in order to reduce, time constraints involved and to allow a large number of respondents. Convenience sampling technique was used in selecting the respondents because this method made use of respondents who were accessible and are ready to be included in the research study. This sampling technique was more appropriate due to the tight schedule of respondents.

Instruments

To ensure soundness and dependability of the instrument, the researchers pilot tested it in LA community Hospital, a hospital whose employees had comparable characteristics as the 37 Military Hospital. The pretesting helped in enlightening vague statements, poorly worded questions that might not be understood by respondents, unclear choices and double-barrelled questions to be taken care of. The result of this exercise helped to adapt the instrument so as to make it appropriate in collecting the desired data. After construction and critical study of the questionnaire, it was passed to the researcher's supervisor for vetting and possible modification of some aspect of the questionnaire. The face validity ensured the technicality of the items in the questionnaire in order to find out its relevance to the study, also the content

validity ensured the technicality of the items in the questionnaire in order to find out if it is relevant to the study, also the content validity of the instrument checked in order to ensure their specification.

For the purpose of this study, the re-test method was used to test reliability of the data collectively. This entails administering the same questionnaire on the same respondents after an interval of three (3) working days. The instrument had some inherent problems. For instance, some of the items were misinterpreted due to poor wording or differential meaning of terms. On the other hand, the researchers ensured that all factors or variables that may influence the validity and reliability of the questionnaire are taken care off.

Data Collection Procedure

The preliminary questionnaire developed was piloted to confirm that the questions were apparent and not vague, such that responses were reliable with the purpose of the study. Not many questions were revealed as a result of non-response from the respondents of pilot study. This was done to advance the dependability and validity of the questionnaire. The questionnaires were self-administered on one-on-one basis by the researchers. The questionnaires were handed to respondents and assured that the questionnaires should take between 20-25 minutes to complete and that the finished questionnaires were collected within five (5) working days.

Retention Rate

A total of forty (40) questionnaires were distributed to the sample population. Out of this, the researches retrieved a total of thirty-seven (37) questionnaires which represents 93% of the distributed questionnaires.

Data Analysis

A Microsoft program, Excel 2016 was used to determine the percentage rate of each answer given by the respondents. The researchers coded the data manually and the analysis were done using descriptive statistics tools to make the analysis easier and understandable to all hence, the researchers critically analyzed the collected data in order to make a concrete decision. The primary data was analyzed by the use of frequency distribution tables to interpret the data collected. This was used to present results and validate the findings of the study. Data collection was done between the period of 24th to 29th March, 2019. The analyzed data was then discussed within the context of the literature reviewed.

CHAPTER FOUR

RESULTS AND DISCUSSION

Introduction

The focus of this chapter is to analyze the data collected and examine the findings in the light of the research questions of the study. The research examines the adoption of information Technology in Ghanaian hospitals with reference to hospital.

Table 1 Gender Distribution of Respondents

Gender	Frequency	Percentage
Male	15	41
Female	22	59
Total	37	100

Source: Field Study, 2019

Table 1 indicates that fifteen (15) which represent 41% of the sampled population were males and the other twenty-two (22) which also represents 59% were females from the total respondents of thirty-seven (37). This implies that majority of the staff were females. This indicates that the hospital industry is dominated by females.

Table 2 Age Distribution of Respondents

Age	Frequency	Percentage
20-29yrs	8	22
30-39yrs	12	32
40 years and above	17	46
Total	37	100

Source: Field Study, 2019

Table 2 observed that, seventeen (17) employees out of thirty-seven (37) respondents representing 46% were above 40 years. Twelve (12) respondents representing 32% were between the ages of 30-39 years, eight (8) respondents representing 22% were between the ages of 20-29 years. This indicates that the majority of the employees are matured. This indicates that most of the respondents were relatively older and more experienced. This could also indicate that the hospital fails to employ or retain younger workforce.

Table 3 Marital Status of Respondents

Marital Status	Frequency	Percentage
Single	15	41
Married	11	30
Divorced	6	16
Widowed	5	13
Total	37	100

Source: Field Study, 2019

Table 3 shows the marital status of the respondents. Fifteen (15) respondents representing 41% of the respondents were single, eleven (11) which represent 30% of respondents were married, six (6) which represents 16% of respondents were divorced while four (5) of the respondents

representing 13% were widowed. This implies that most of the respondents were single and likely to invest their skills and energy in the job.

Table 4 Academic Qualification of the Respondents

Qualification	Frequency	Percentage
Undergraduate	21	58
Graduate	8	21
Other Qualifications	8	21
Total	37	100

Source: Field Study, 2019

Table 4 illustrates that twenty-one (21) respondents representing 58% of the respondents were undergraduates. However, eight (8) of the respondents representing 21% were Graduate degree holders while eight (8) respondents representing 21% had obtained other academic qualifications did not specify what qualifications they held. The results show that majority of the respondents were first degree holders and are more likely to know the performance of the hospital.

Table 5 Years of Service of the Respondents.

Category	Frequency	Percentage
1-5 years	7	19
6-10 years	11	30
11-15 years	14	38
16 and Above	5	13
Total	37	100

Source: Field Study, 2019

Table 5 shows that seven (7) respondents representing 19% of the respondents have served between 1-5 years. Eleven (11) of the respondents

representing 30% have served between 6-10 years, fourteen (14) respondents representing 38% had served the hospital between 11-15 years while five (5) respondents representing 13% of the respondents have served the hospital above 16 years. This shows that most of the employees have been with the hospital consistently and are more likely to know about the hospital's performance.

Table 6 Awareness of ICT

Category	Frequency	Percentage
Excellent Knowledge	2	6
Good knowledge	29	78
Average Knowledge	3	8
Little Knowledge	3	8
Total	37	100

Source: Field Study, 2019

Table 6 shows level of knowledge or awareness of ICT. Two (2) respondents which represents 6% of the respondents had excellent knowledge, twenty-nine (29) which represents 78% had good knowledge, three (3) representing 8% had average knowledge while three (3) which represents 8% had little knowledge. The outcome shows that majority of the respondent's knowledge in ICT were good. The essence of the respondents having good knowledge in ICT implies that they understand and could use ICT tools to serve patients better.

Table 7 **ICT Policy**

Category	Frequency	Percentage
Yes	37	100
No	0	0
Total	37	100

Source: Field Study, 2019

Table 7 indicates that twenty (37) respondents representing 100% said that the hospital had an IT policy as compared to zero (0) representing 0% said the hospital had no IT policy. The results show that the hospital had an IT policy.

Table 8 **ICT Application**

Category	Frequency	Percentage
Yes	0	0
No	100	100
Total	37	100

Source Field: Study, 2019

Table 8 indicates whether the hospital currently have no ICT applications that can be easily adopted to provide access to patients and clients to undergo check-up's online. Zero (0) respondents representing 0% of the respondents agreed, thirty-seven (37) respondents representing 100% respondents disagreed to the hospital having ICT application devices. This affirms that the hospital does not have an ICT application that can easily be adopted to provide access to patients and clients to undergo check-up's, it was found that the hospital does not have the adequate facilities to enable clients to undergo such programme. Besides, it was gathered that most of the

respondents does not support the idea of clients using ICT applications to undergo check-ups online.

Table 9 ICT Application

Category	Frequency	Percentage
Yes	37	100
No	0	0
Total	37	100

Source: Field Study, 2019

Table 9 shows to whether the hospital possess an online presence showing the hospital's products and services most of the respondents indicated that the hospital has a website showing products and services to patients and clients. This was agreed to by thirty-seven (37) respondents representing 100% of respondents as compared to zero (0) respondents representing 0% who disagreed. The outcome shows that the hospital possesses an online presence showing the hospital's products and services to patients and clients. It was found that the hospital rarely sends bills/charges to patients through emails and also, allows patients to access medication through ICT application. Furthermore, it was discovered that the hospital market products and services to patients and clients who use their website. During the study, it was gathered that 37 Military hospital have excellent infrastructure such as hardware, network services, software application including firewall, antivirus and IT experts whose specialties were into database administration and security as well as employees with special skills in information and communication Technology.

Table 10 **Importance of ICT Implementation**

Category	Frequency	Percentage
Very important	37	100
Not Important	0	0
Do not know	0	0
Total	37	100

Source field Study, 2019

Table 10 indicates how important is the implementation of ICT, thirty-seven (37) respondents representing 100% said it was very important, zero (0) respondents representing 0% mentioned it was not whiles zero (0) which represented 0% were undecided. The results show that the implementation of ICT was very important to the hospital due to the rapidly changing advances in information and communications technologies (ICT) and computer technologies. This has led to the hospital to embrace these changes to achieve a competitive edge in the industry.

Table 11 **Perspective of ICT**

Category	Frequency	Percentage
Opportunity	37	100
Challenge	0	0
Threat	0	0
Total	37	100

Source: Field Study, 2019

Table 11 indicates how the hospital perceives ICT, thirty-seven (37) respondents representing (100%) of the respondents perceived ICT as opportunity, zero (0) respondents which represents 0% saw it as a challenge whiles zero (0) which represents 0% saw it as a threat. The results show that

the hospital perceived information communication Technology as an opportunity to transact and outline products and services to patients and clients.

Table 12 Adoption of ICT

Category	Frequency	Percentage
Agree	37	100
Disagree	0	0
Total	37	100

Source: Field Study, 2019

Table 12 indicates whether the adoption of ICT has improved the fortune of the hospital, thirty-seven (37) which represents 100% of the respondents agreed, zero (0) which represents 0% disagreed. The results show that the adoption of ICT has improved the fortunes of the hospital.

Table 13 Introduction of ICT

Category	Frequency	Percentage
Agreed	37	100
Disagree	0	0
Total	37	100

Source: Field Study, 2019

Table 13 depicts whether ICT has impacted positively on the overall performance of the hospital, thirty-seven (37) which represents 100% of the respondents agreed, zero (0) which represents 0% disagreed. The outcome shows that information communication Technology has impacted positively on the overall performance of the hospital.

Table 14 Risks Involved in ICT		
Category	Frequency	Percentage
High	0	0
Low	37	100
Total	37	100

Source: Field Study, 2019

Table 14 is in response to how the hospital assess the risk involved in ICT scheme, zero (0) respondents rate the threat of assessment high, thirty (37) respondents representing 100%, rate the threat of assessment low. The outcome shows that the hospital has a low incidence of threat scheme. On the question of the greatest challenges that hinder the adaption of ICT, it was found that the lack of resources, knowledge, skill, security concerned and firm readiness were some of the challenges identified by the respondents.

CHAPTER FIVE

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Introduction

The purpose of this chapter is to round off the study with summary of findings, conclusion and recommendations. The conclusions are made from the analysis and the objectives of the research.

The study was carried out in order to examine the adoption of information technology at hospitals. The research design adopted for the study was survey design the choice of the study design was motivated by the fact that it best describes the opinion, personal characteristics, perceptions, preferences and attitude of respondents. The respondents of the study were employees of employees of the 37 Military Hospital in Accra. The sample size of the study was a total of forty (40) respondents. This number was used to ensure that there was a general presentation of the population. Convenience sampling technique was used in selecting the respondents because this method made use of respondents who were accessible and are ready to be included in the research study. Microsoft Excel 2016 was used to determine the percentage rate of each answer given by the respondents. The researchers coded the data manually and the analyses were done using descriptive statistics tools to make the analysis easier and understandable to all during the study, it was revealed that most of the respondent's knowledge in ICT was good which implied that they understood and could use ICT tools to serve patients better. It was found

that the hospital had an ICT policy to aid their work. The results showed that the hospital possesses an online presence showing the hospital's products and services to patients and clients. The data confirmed that, the implementation of ICT was very important to the hospital due to the rapidly changing advances in Information and Communication Technologies (ICT) and computer technologies. This has enabled the hospital to embrace these changes to achieve a competitive edge in the industry. The findings clearly revealed that the hospital perceived Information Communication Technology as an opportunity to transact and outline products and services to patients and clients. The results showed that the introduction of Information and Communication Technology (ICT) has improved patient satisfaction. It was found that the adoption of ICT has improved the fortunes of the hospital and also, impacted positively on the overall performance of the hospital finally, it was discovered that the lack of resources, knowledge, skill, security concerns and firm readiness were some of the challenges identified by the respondents.

Conclusions

From the gathered data, it can be concluded that employee's respondent's knowledge in ICT was excellent which implied that they understand and could Information communication Technology (ICT) tools to serve patients better, it can also be concluded that the hospital had an ICT policy to aid their work. Other conclusions that can be gathered from the study include the following:

- a. The hospital possesses an online presence showing the hospital's products and services to patents and clients.

- b. The hospital perceived Information and Communication Technology as an opportunity to transact and outline products and services to patients and clients.
- c. The introduction of ICT has improved patient satisfaction.
- d. The adoption of ICT has improved the fortunes of the hospital and also, impacted positively on the overall performance of the hospital.
- e. The lack of resources, knowledge, skill, security concerns and firm readiness were some of the challenges identified by the respondents

The above also meant that the researchers can confirmed that the implementation of ICT was very important to the hospital due to the rapidly changing advances in information and communication technologies (ICT) and computer technologies. This has enabled the hospital to embrace these changes to achieve a competitive edge in the industry.

Recommendations

In view of the objectives set and findings thereof, the following recommendations have been made:

It is recommended that the hospital must provide adequate resources to address security concerns, employee's knowledge in the adaption of ICT. Again, it is recommended that the hospital must change patients and clients about ICT adaption of products and services. They should not be content but should continue strive to make improvements in all their operations and governance in order to achieve the highest quality service that is demanded from patients and clients.

It is recommended that, in order to succeed online, hospital products should be offered at discount and be better than traditional offline options. Without this incentive, patients and clients would not see the need of conducting insurance business online.

It is recommended that the 37 Military hospital need to build patients awareness and understanding of ICT as most people in the country perceive ICT to be for the affluent.

Finally, it is recommended that the government of Ghana needs to reduce tax and duty on ICT equipment's to make technologies accessible and affordable to all hospitals in Ghana.

Recommendations for future Research

It is recommended that the same study be conducted performed with larger sample size in the future. A larger sample would enable researchers to examine how hospitals should provide adequate resources to address security concerns, employee's knowledge in the adaption of information communication Technology. By this, hospitals and employees could enhance their skills and knowledge on Information Communication Technology and hence overall organizational improvement.

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APPENDICES

APPENDIX A:
QUESTIONNAIRE FOR THE 37 Military Hospital

Introduction

The purpose for this study is to fulfill the requirement for the award of Bachelor of Business Administration Degree in Business Information System (BIS), at the Department of Information and Communication Technology, Zenith College. The study solicits for data in order to examine "the Adoption of Information Technology in Ghanaian hospitals with reference to 37 Military hospital". You are assured that responses given will be treated with confidentiality and will be used solely for academic purposes. Kindly answer the questions as objectively as possible.

Thank you.

Section A: Demographic data

Instruction: Please tick [☐] in the appropriate box provided to indicate your answers.

1. Gender?

Male [☐] Female [☐]

2. Age bracket?

20-29 yrs [☐] 30-39 yrs [☐] 40 yrs and Above [☐]

3. Marital Status?

Single [☐] Married [☐] Divorced [☐] Widowed [☐]

4. Academic qualification

Undergraduate [☐] Graduate [☐] Other Qualifications [☐]

5. Length of service: How long have you been in your current organization?

1-5yrs [☐] 6-10yrs [☐] 11-15yrs [☐] 16yrs and above [☐]

Section B: Impact on Information and Communication Technology

6. What is your level of ICT knowledge?

Excellent [☐] Good [☐] Average [☐] Little/Poor [☐]

7. Does the hospital have ICT policy?

Yes [☐] No [☐]

8. Does the hospital currently have an ICT application that can be easily adopted to provide access to patients and clients to undergo checkups online?

Yes [☐] No [☐]

9. Does the hospital possess an online presence showing the hospital's products and services?

Yes [☐] No [☐]

10. How frequently does the hospital use the following business applications and technologies?

	ALWAYS	RARELY	NEVER
E-MAIL			
a. Market Hospital Products and Services			
b. Send Bills/Charges to Clients			
c. Allow Patients to Access Medication			
WORLD WIDE WEB (WWW)			
a. Display Hospital Products and Services			
b. Obtain Information About Patients			
c. Allow Patients to Purchase Drugs			

11. To what extent is your hospital equipped to implement ICT in your view of the following?

	GOOD	POOR
INFRASTRUCTURE (HARDWARE, NETWORK ETC.)		
SOFTWARE APPLICATIONS (FIREWALLS, ANTIVIRUSES ETC)		
IT EXPERTS (DATABASE, , ADMINISTRATION, SECURITY ETC.		
EMPLOYEES SKILLED IN IT		

Section C: Factors affecting ICT Adoption

12. How important is the implementation of ICT to your hospital?

Very important [] Not important [] Do not know []

13. For your hospital, ICT is/an?

Opportunity [] Challenge [] Threat []

14. The adoption of ICT has improved the fortune of the hospital?

Agree [] Disagree []

15. The introduction of ICT has improved patient satisfaction?

Agree [] Disagree []

16. 37 Military hospital ICT has impacted on its overall performance?

Agree [] Disagree []

Section D: Challenges that hinder the adoption of ICT

17. How can you access the incidence risk of 37 Military hospital ICT scheme?

High [] Low []